# SDCHA1V80

## Automotive grade semi-shielded power inductors




### Applications

- LED lighting
- Advanced driver assistance systems (ADAS)
- Adaptive cruise control (ACC)
- Collision avoidance
- Infotainment and cluster electronics
- Electronic control unit (ECU)

### Product features

- AEC-Q200 qualified
- High current carrying capacity
- High power density, low core losses
- Magnetically semi-shielded
- Inductance range from 1 µH to 100 µH
- Current range from 1.2 A to 8.5 A
- 8.3 mm x 8.3 mm surface mount package in a maximum 4.0 mm height
- NiZn ferrite magnetic material
- Moisture sensitivity level (MSL): 1

### Environmental compliance and general specifications

- Storage temperature range (component): -55 °C to +125 °C
- Operating temperature range: -55 °C to +125 °C (ambient plus self-temperature rise)
- Solder reflow temperature: J-STD-020 (latest revision) compliant

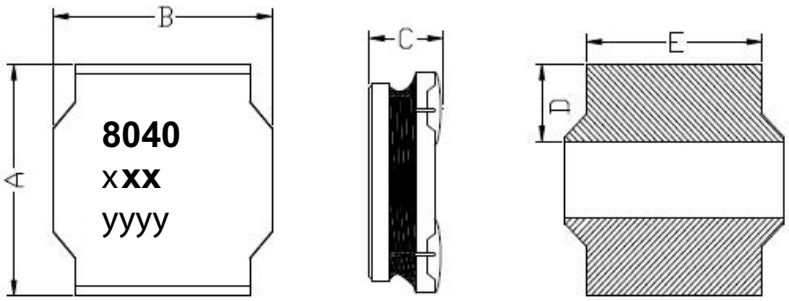
Powering Business Worldwide

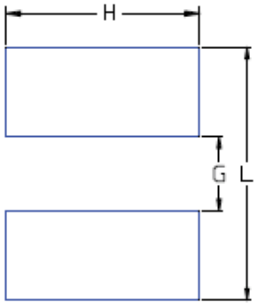
## Product specifications

| Part number[5] | OCL[1] (µH) | Tolerance | FLL[2] (µH) minimum | $I_{rms}$[3] (A) | $I_{sat}$[4] (A) | DCR (mΩ) ±20% @ +25 °C | SRF (MHz) typical |
|---|---|---|---|---|---|---|---|
| SDCHA1V8040-1R0-R | 1.0 | ±20% | 0.56 | 8.5 | 13.8 | 8.2 | 85 |
| SDCHA1V8040-1R5-R | 1.5 | ±20% | 0.84 | 8.0 | 11.5 | 10 | 66 |
| SDCHA1V8040-2R2-R | 2.2 | ±20% | 1.23 | 7.4 | 9.8 | 11.5 | 57 |
| SDCHA1V8040-3R3-R | 3.3 | ±20% | 1.85 | 6.6 | 8.0 | 15 | 48 |
| SDCHA1V8040-4R7-R | 4.7 | ±20% | 2.63 | 5.8 | 6.7 | 19.5 | 38 |
| SDCHA1V8040-5R6-R | 5.6 | ±20% | 3.14 | 5.4 | 6.2 | 22 | 33 |
| SDCHA1V8040-6R8-R | 6.8 | ±20% | 3.81 | 5.1 | 5.6 | 25 | 29 |
| SDCHA1V8040-100-R | 10 | ±20% | 5.60 | 4.6 | 5.0 | 33 | 25 |
| SDCHA1V8040-150-R | 15 | ±20% | 8.40 | 3.6 | 4.0 | 50 | 20 |
| SDCHA1V8040-220-R | 22 | ±20% | 12.32 | 2.9 | 3.1 | 73 | 18 |
| SDCHA1V8040-330-R | 33 | ±20% | 18.48 | 2.3 | 2.6 | 100 | 15 |
| SDCHA1V8040-470-R | 47 | ±20% | 26.32 | 2.0 | 2.2 | 135 | 12 |
| SDCHA1V8040-560-R | 56 | ±20% | 31.36 | 1.75 | 1.9 | 160 | 10.5 |
| SDCHA1V8040-680-R | 68 | ±20% | 38.08 | 1.65 | 1.75 | 205 | 9.5 |
| SDCHA1V8040-820-R | 82 | ±20% | 45.92 | 1.4 | 1.6 | 230 | 8.5 |
| SDCHA1V8040-101-R | 100 | ±20% | 56 | 1.2 | 1.45 | 300 | 8.0 |

1. Open circuit inductance (OCL) test parameters: 100 kHz, 1.0 Vrms, 0.0 Adc, +25 °C
2. Full load inductance (FLL) test parameters: 100 kHz, 1.0 Vrms, $I_{sat}$, +25 °C
3. $I_{rms}$: DC current for an approximate temperature rise of 40 °C without core loss. Derating is necessary for AC currents. PCB layout, trace thickness and width, air-flow, and proximity of other heat generating components will affect the temperature rise. It is recommended that the temperature of the part not exceed +125 °C under worst case operating conditions verified in the end application.
4. $I_{sat}$ : Peak current for approximately 30% maximum rolloff @ +25 °C
5. Part number definition: SDCHA1V8040-xxx-R
   SDCHA1V8040 = Product code and size
   xxx= Inductance value in µH, R=decimal point, If no R is present last digit indicates number of zeros
   -R suffix = RoHS compliant
6. Rated operating voltage (across inductor) 20 V ref.

**Dimensions-mm**




Recommended PCB Layout



Schematic

| Dimension | SDCHA1V8040-xxx-R |
|---|---|
| A | 8.0 ± 0.3 |
| B | 8.0 ± 0.3 |
| C | 3.7 ± 0.3 |
| D | 2.4 ± 0.3 |
| E | 6.3 ± 0.3 |
| L | 8.5 |
| G | 2.8 |
| H | 6.6 |

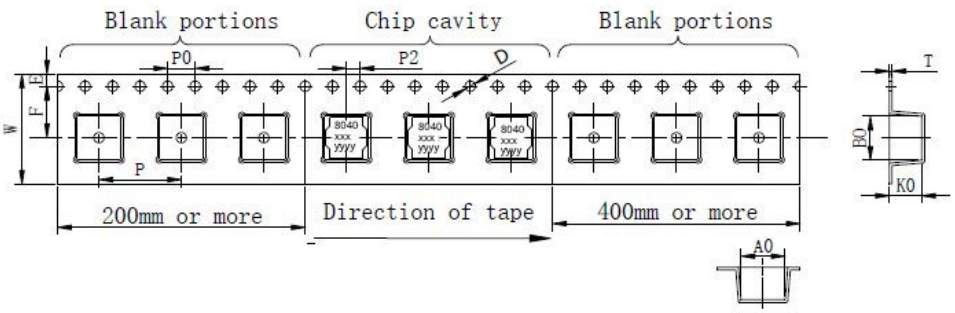
Part marking: 8040, xxx= inductance value in uH, R= decimal point. If no R is present then last character equals number of zeros, yyyy= lot code
All soldering surfaces to be coplanar within 0.1 millimeters
PCB layout reference only
Recommend solder paste thickness at 0.15 mm and above
Traces or vias underneath the inductor is not recommended

**Packaging information- mm**

**SDCHA1V8040**

Supplied in tape and reel packaging, 1000 parts per 13" diameter reel (EIA-481 compliant)

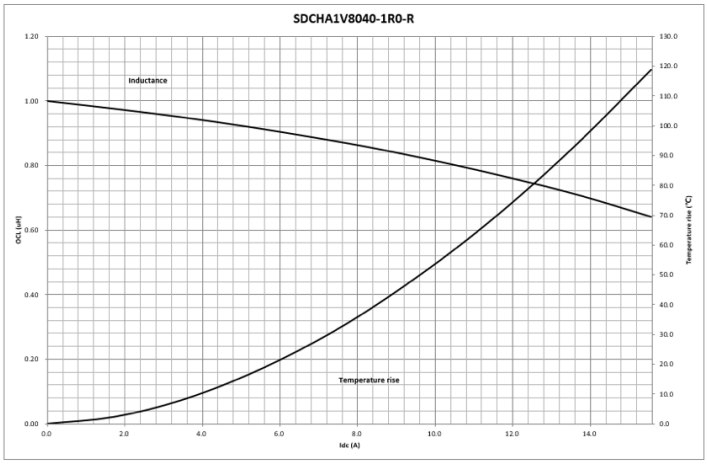
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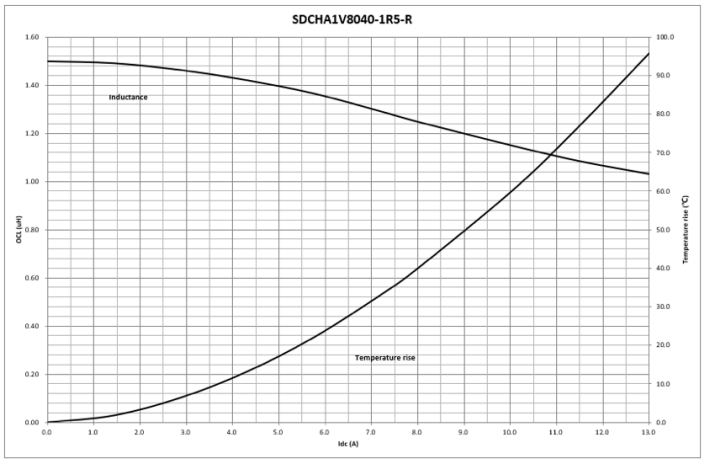


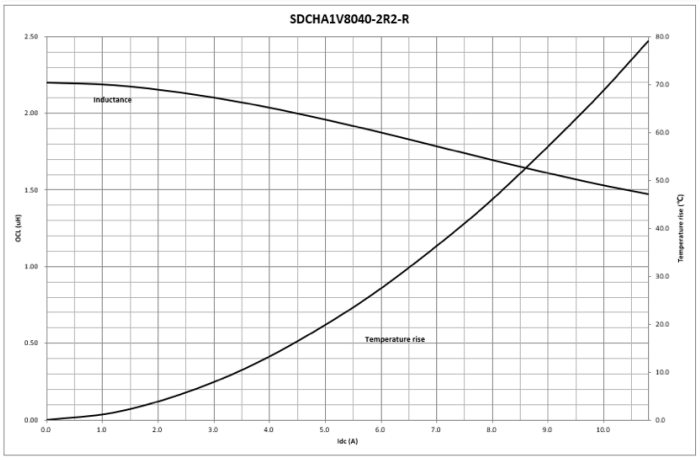

| Dimension | Value |
|---|---|
| W | 16.0 ± 0.3 |
| F | 7.5 ± 0.1 |
| E | 1.75 ± 0.1 |
| P0 | 4.0 ± 0.1 |
| P | 12.0 ± 0.1 |
| P2 | 2.0 ± 0.1 |
| D | 1.5 ± 0.1 |
| A0 | 8.4 ± 0.1 |
| B0 | 8.4 ± 0.1 |
| K0 | 4.3 ± 0.1 |
| T | 0.4 ± 0.1 |

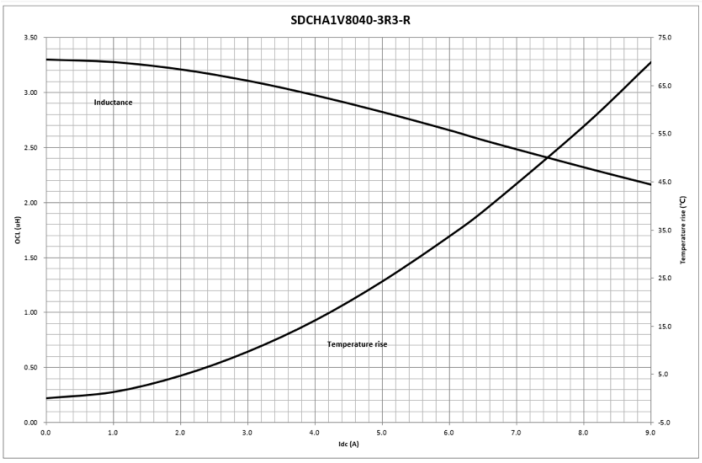
**Inductance and temperature rise vs current**

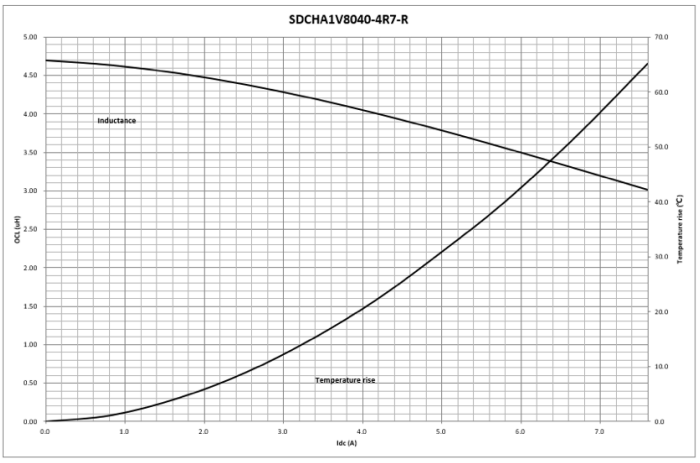
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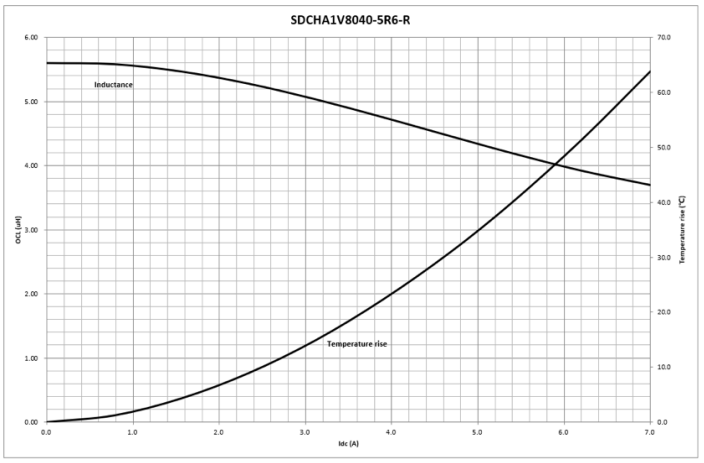


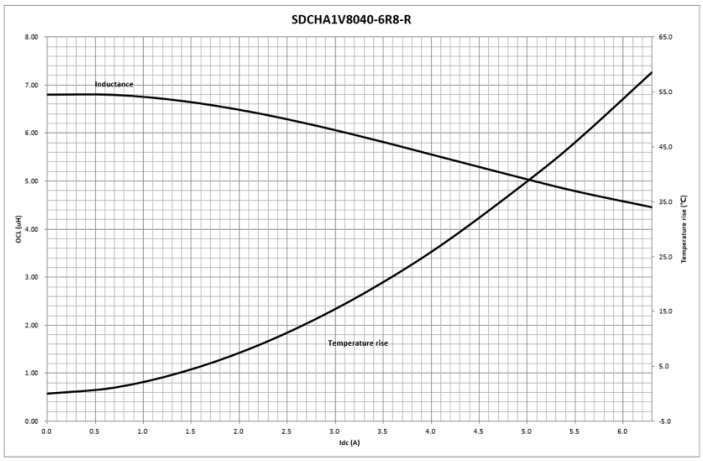



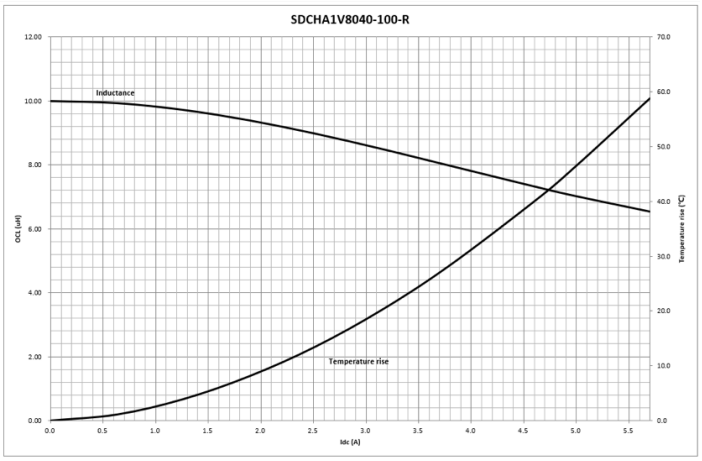



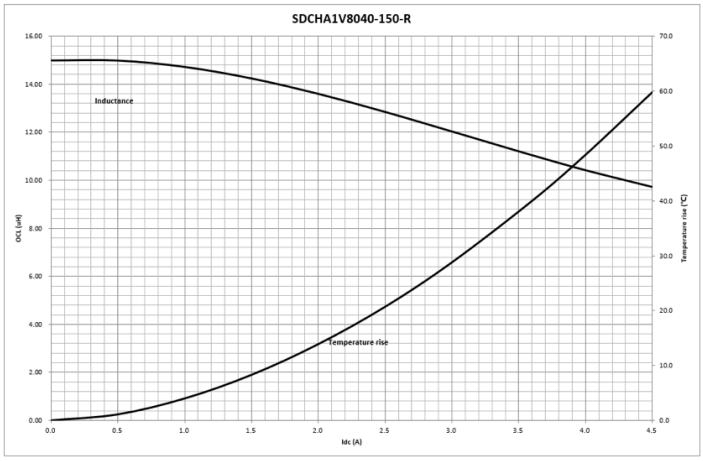



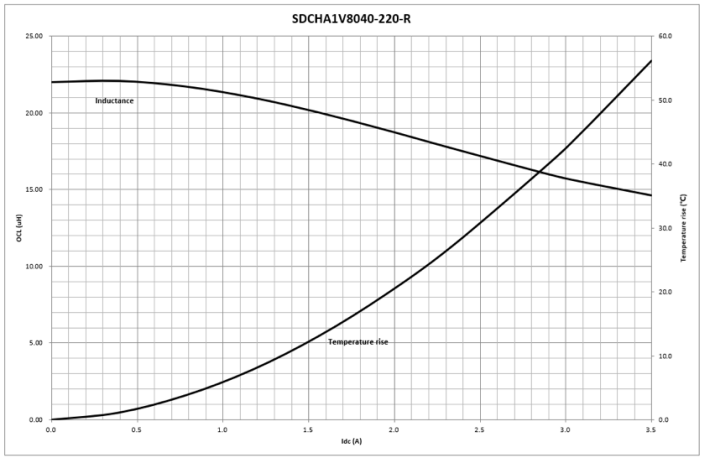



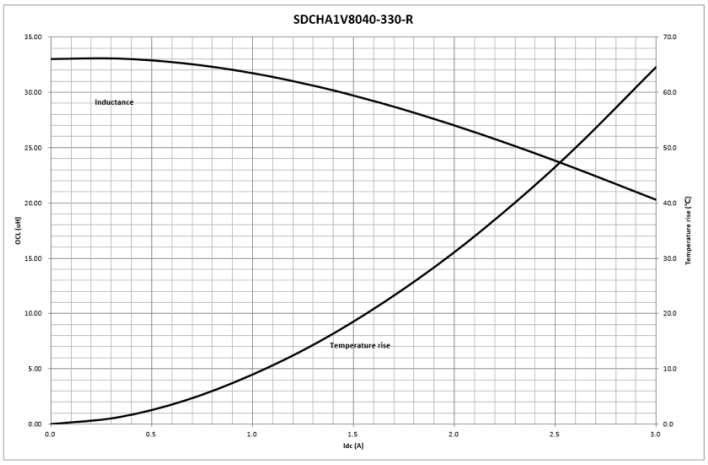
**Inductance and temperature rise vs current**

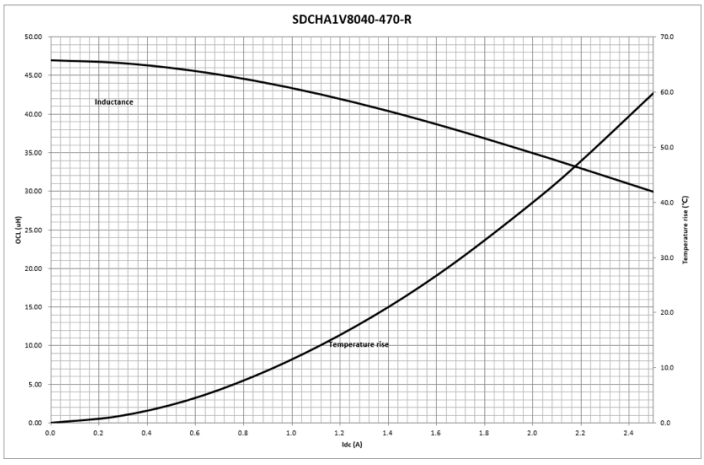


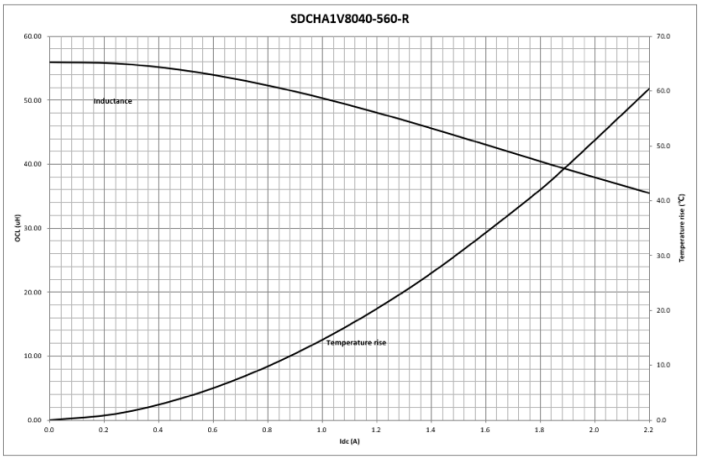



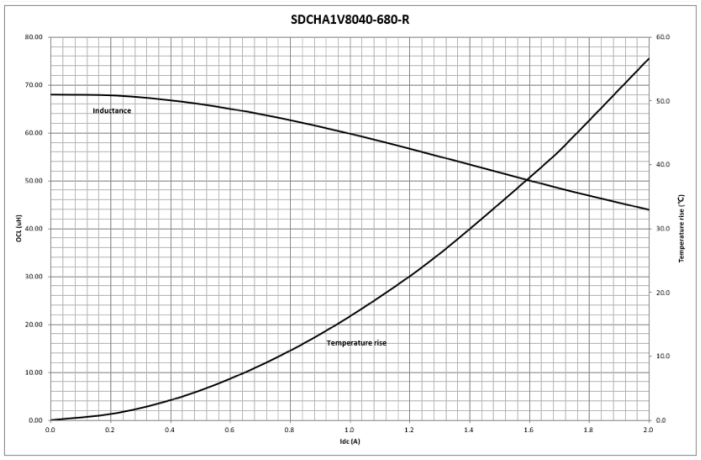



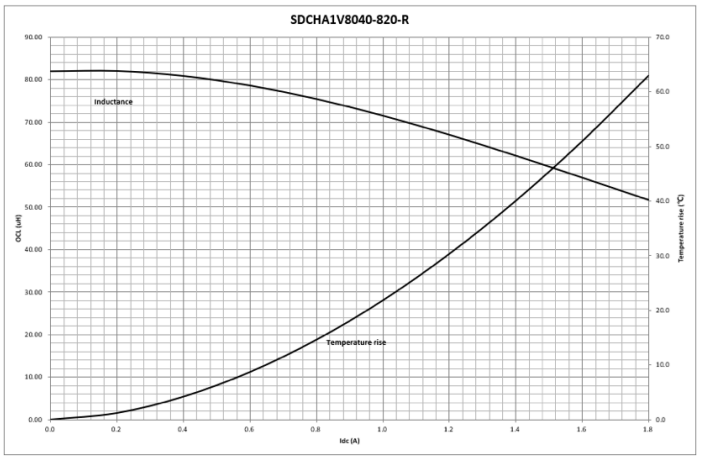



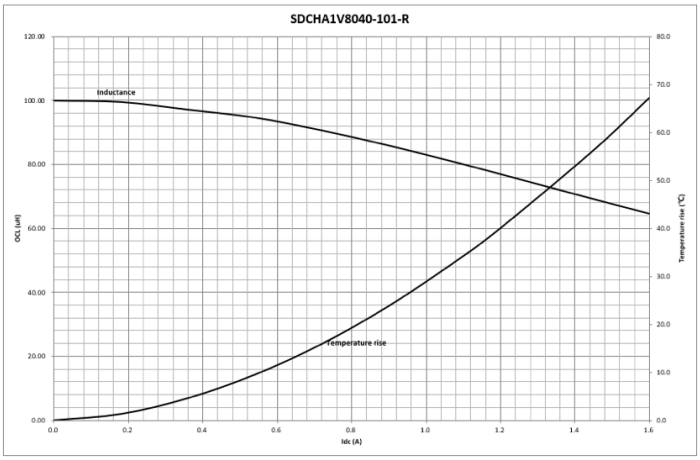



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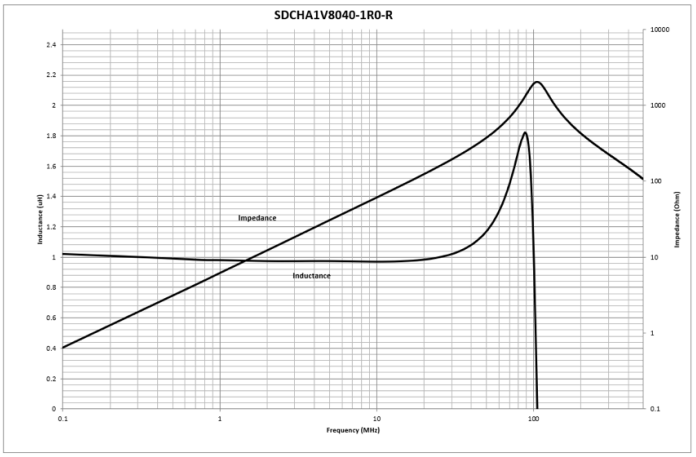


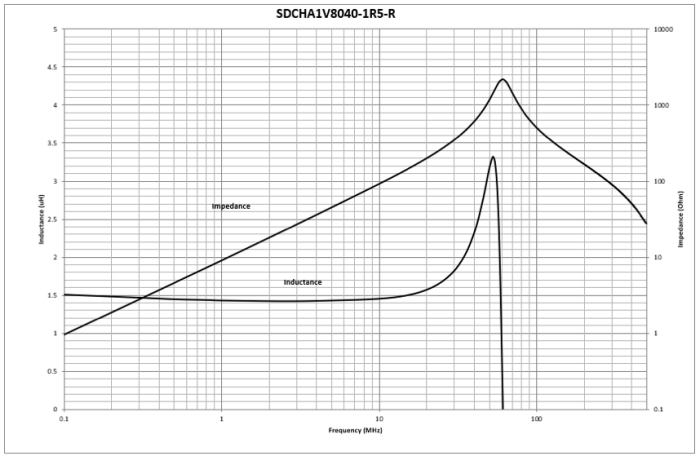



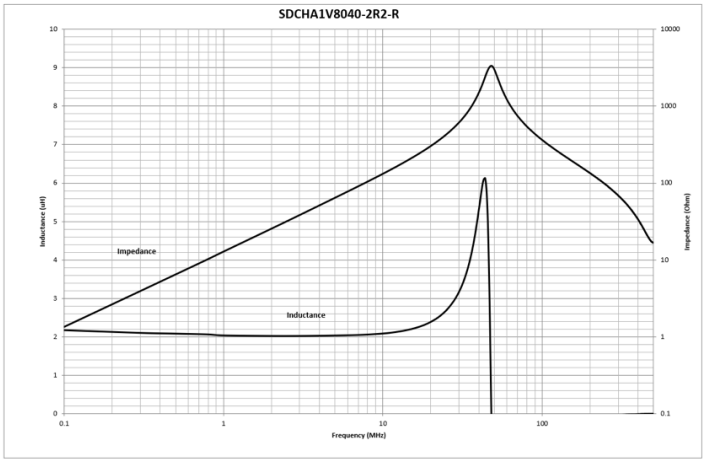



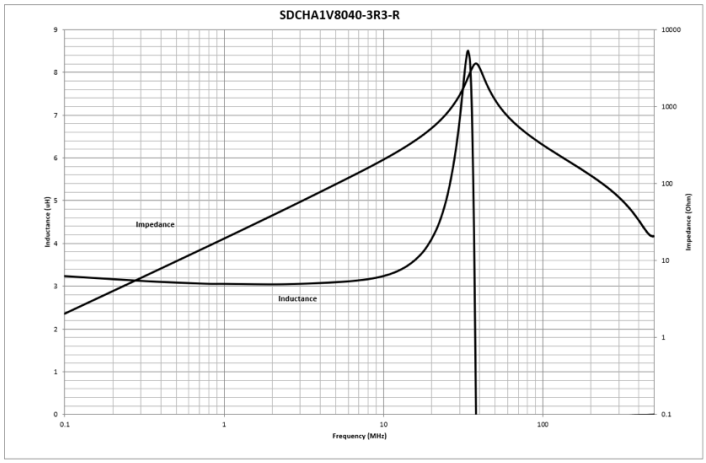



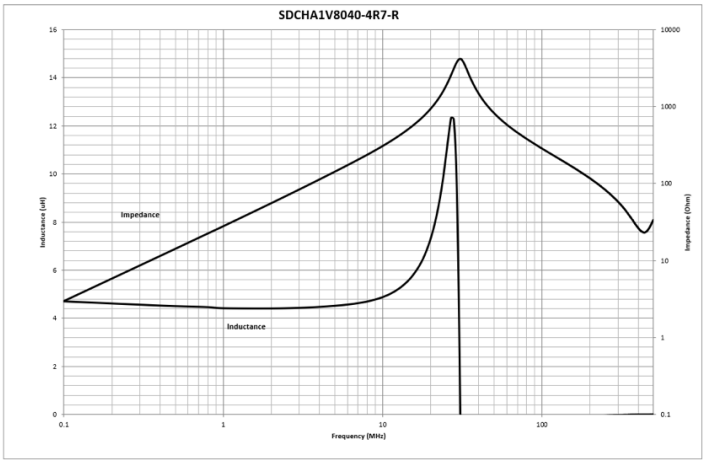
**Inductance and impedance vs. frequency curve**

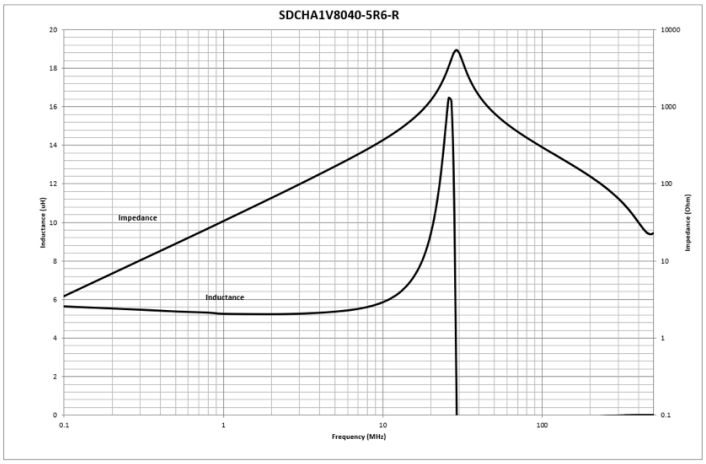


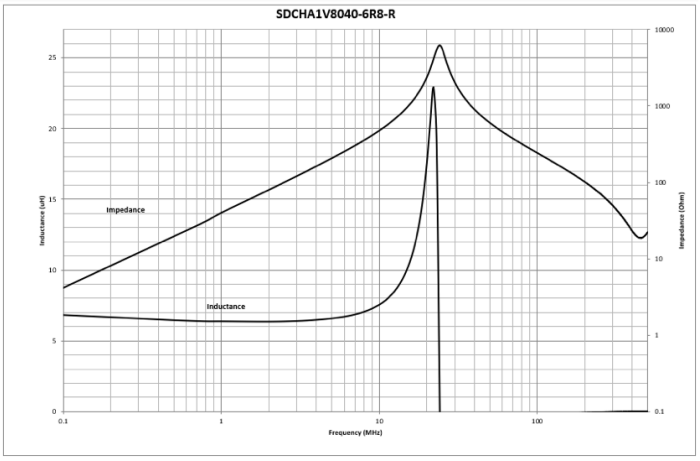



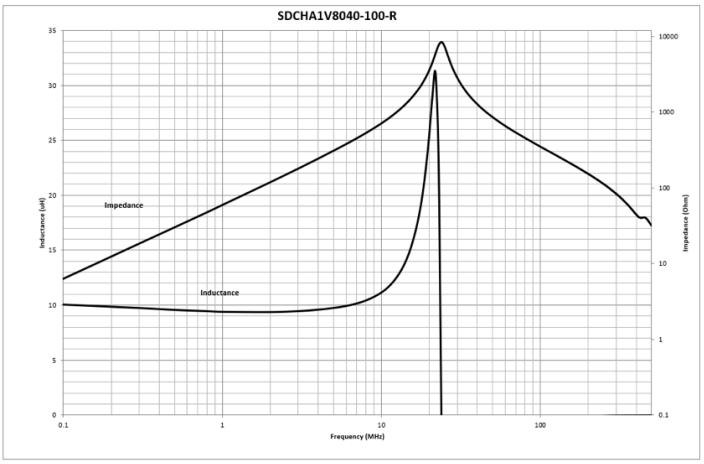



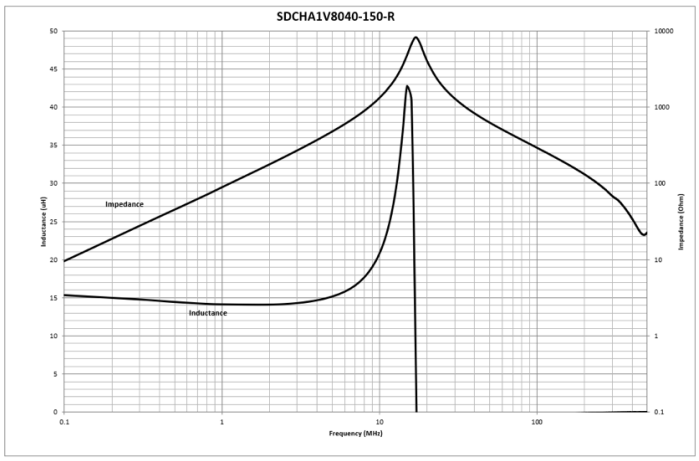



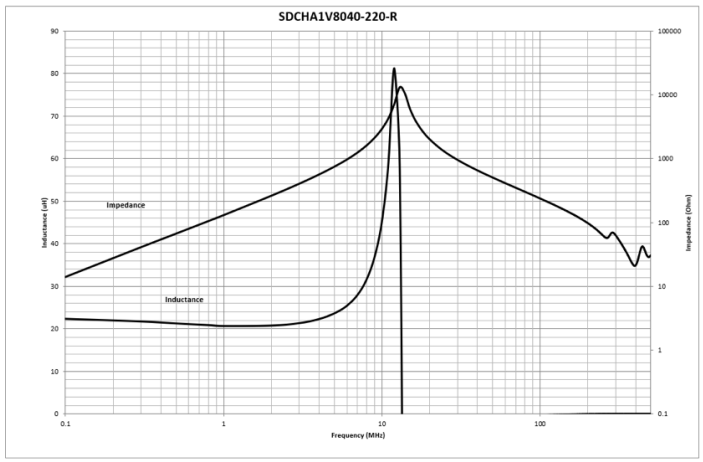



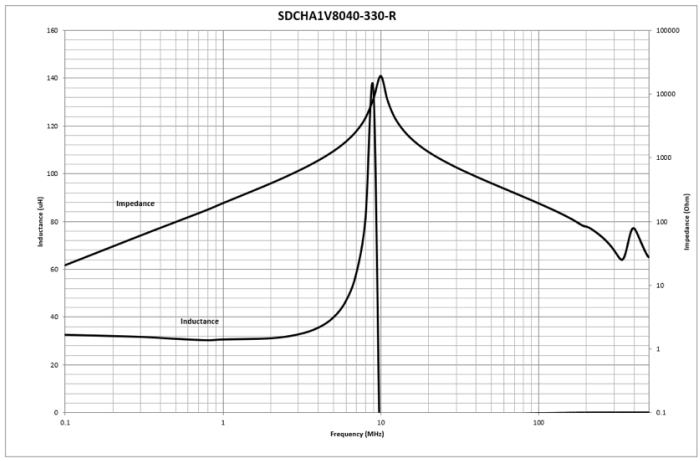
**Inductance and impedance vs. frequency curve**

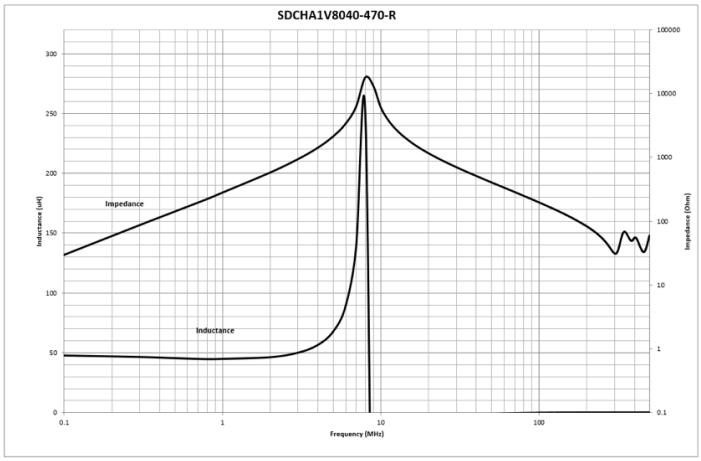


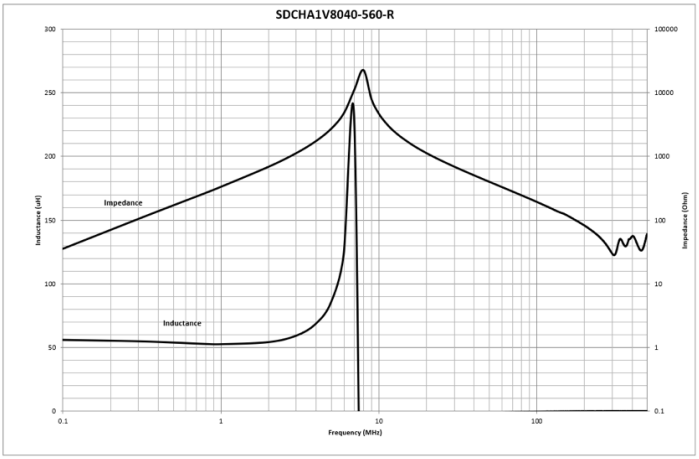



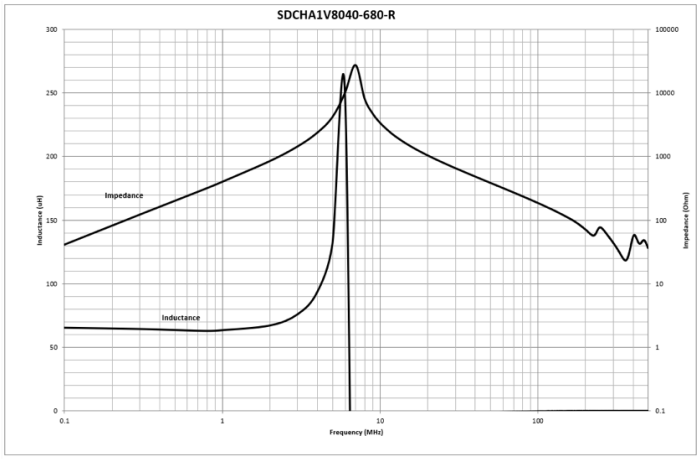



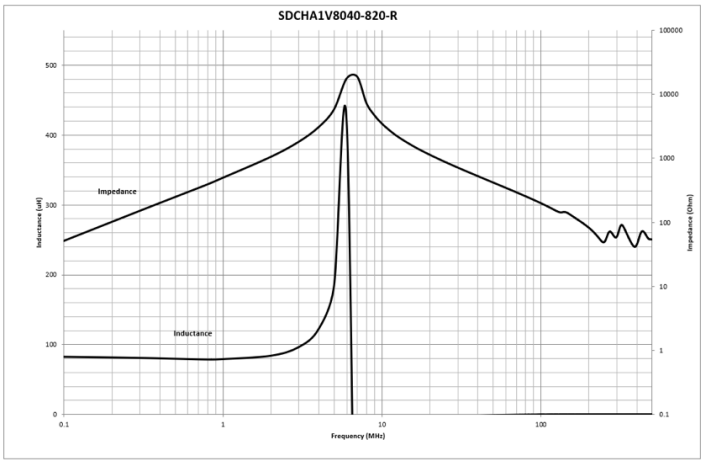



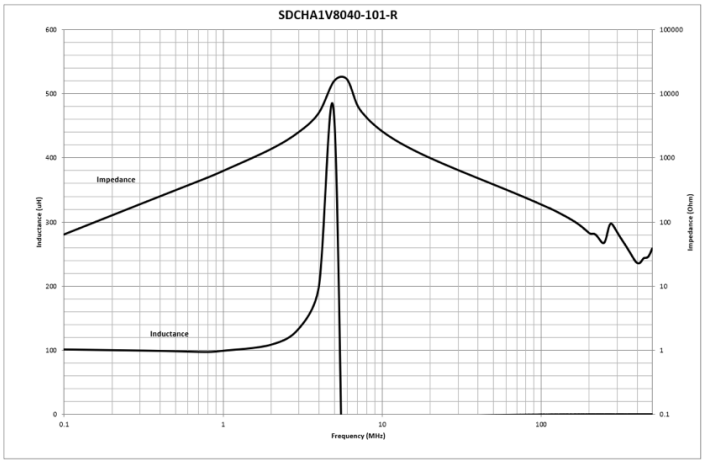



**Inductance and impedance vs. frequency curve**



## Solder reflow profile

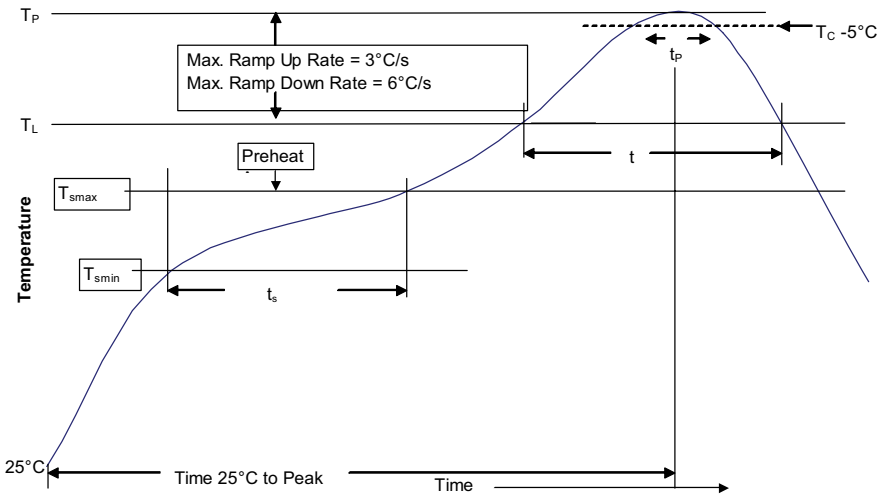



**Table 1 - Standard SnPb solder ($T_C$)**

| Package thickness | Volume mm3 <350 | Volume mm3 ≥350 |
|---|---|---|
| <2.5 mm) | 235 °C | 220 °C |
| ≥2.5 mm | 220 °C | 220 °C |

**Table 2 - Lead (Pb) free solder ($T_C$)**

| Package thickness | Volume mm³ <350 | Volume mm³ 350 - 2000 | Volume mm³ >2000 |
|---|---|---|---|
| <1.6 mm | 260 °C | 260 °C | 260 °C |
| 1.6 – 2.5 mm | 260 °C | 250 °C | 245 °C |
| >2.5 mm | 250 °C | 245 °C | 245 °C |

## Reference J-STD-020

| Profile feature | | Standard SnPb solder | Lead (Pb) free solder |
|---|---|---|---|
| Preheat and soak | • Temperature min. ($T_{smin}$) | 100 °C | 150 °C |
| | • Temperature max. ($T_{smax}$) | 150 °C | 200 °C |
| | • Time ($T_{smin}$ to $T_{smax}$) ($t_s$) | 60-120 seconds | 60-120 seconds |
| Ramp up rate $T_L$ to $T_p$ | | 3 °C/ second max. | 3 °C/ second max. |
| Liquidous temperature ($T_L$)<br>Time ($t_L$) maintained above $T_L$ | | 183 °C<br>60-150 seconds | 217 °C<br>60-150 seconds |
| Peak package body temperature ($T_P$)* | | Table 1 | Table 2 |
| Time ($t_p$)* within 5 °C of the specified classification temperature ($T_C$) | | 20 seconds* | 30 seconds* |
| Ramp-down rate ($T_p$ to $T_L$) | | 6 °C/ second max. | 6 °C/ second max. |
| Time 25 °C to peak temperature | | 6 minutes max. | 8 minutes max. |

* Tolerance for peak profile temperature ($T_p$) is defined as a supplier minimum and a user maximum.

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